

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of BSc (Biotechnology) Examination May, 2018

BT603 Basic Animal Biotechnology

Date: 04/05/18 Day: Friday Time: 1.30 p.m. To 3.30 p.m.

Maximum Marks: 10

MCQ

[illegible]

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.	10
1.	The technique used in animal biotechnology for the rapid multiplication and production of animals with a desirable genotype is (i) protoplast fusion and embryo transfer (ii) IVF and embryo transfer (iii) hybrid selection and embryo transfer (iv) None of these	
2.	DMSO and HES are used as _____ in animal tissue culture. (i) cryoprotectants (ii) antibiotics (iii) antifungal agents (iv) None of these	
3.	The cell culture environment consist of _____ (i) the buffering system (ii) media and its components (iii) the plastic substrate for cell attachment (iv) All of these	

4.	The type of contamination not observed under standard inverted microscope is____ (i) mycoplasma (ii) bacterial (iii) fungal (iv) All of these	
5.	Reduced serum requirement and anchorage independence is the feature of continuous cell lines. (True/False)	
6.	The production of a glassy state during cryopreservation is defined by the viscosity reaching a sufficiently high value to behave like a solid but without any crystallization is called intracellular vitrification. (True/False)	
7.	Who contributed to the successful beginning of animal tissue culture technique in 1907 using frog tissue? (i) Hardy Weinberg (ii) Stevensen (iii) Ross Harrisson (iv) none of these	
8.	Finite Cell lines with limited life span need to undergo _____ several times before they become senescent. (i) passaging (ii) modification (iii) toxicity testing (iv) none of these	
9.	Which of the following is not a survival assay?	
	(i) MTT assay (ii) inflammation assay (iii) clonogenic assay (iv) none of these	
10.	Dolly, a domestic sheep, was the first mammal to be cloned from an adult germinal cell. (True/False)	

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Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed		10
	Answer any five of the following:	
1.	Discuss various applications of Transgenic animals in brief.	
2.	Write down the morphological sign and symptoms of bacterial and fungal contamination in cell cultures.	
3.	How is electroporation method of gene transfer different from gene gun method? Explain.	
4.	Give labeled diagrammatic representation of a cryocontainer.	
5.	Write a short note on CO ₂ incubators used in animal tissue culture labs.	
6.	What is IVF? Write the advantages of this method.	
SECTION - II		
Q-III Answer the following questions as directed		15
	Answer any three of the following:	
1.	Explain retroviral mediated gene transfer in detail. Discuss about the merits and demerits of viral mediated gene transfer technology.	
2.	Differentiate cytotoxicity/viability and survival assays. Discuss one assay each of viability and survival types.	
3.	Write down several aseptic techniques aimed to exclude contamination in cell culture. Add a note on Mycoplasma contamination and its detection methods.	
4.	What are the objectives of gene transfer in animals? Write down the steps of gene transfer and discuss ES cell method as a strategy for gene transfer in animals.	